Fermionic quantum field theories as probabilistic cellular automata in three dimensions

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Wetterich has shown that certain probabilistic cellular automata in $1+1$ dimensions are exactly equivalent to fermionic quantum field theories, and posed the construction of three-dimensional examples as an open problem. We solve the free-fermion sector of this problem under the strictest reading: single-fermion carriers, one homogeneous local invertible update rule, no composite transport. We first prove the obstructions that made the problem hard: ballistic one-particle transport admits only finite velocity sets (no cone); Abelian sign gauges cannot repair this; and two-component internal states admit isotropic nodes only at isolated fine-tuned points, every one frequency locked and without a dial. The solution is a layered automaton ("coincube") whose conversion blocks form the real quaternion representation of $\mathrm{Cl}(0,3)$, controlled by autonomous binary environment fields. We exhibit an explicit complex structure with respect to which the step operator is exactly unitary, prove that the standard spectral analysis is its faithful complexification via an explicit intertwiner, extract the local Grassmann action mechanically in exact arithmetic, and verify the fermionic lift including all sign coherences. By gated measurement on the quenched automaton, the single-carrier excitation on a clustering product vacuum is an isotropic Weyl fermion: slope ratios equal 1 within errors, a factorized dispersion is excluded in every measured channel (weakest channel 9.7 error-bar widths), and the residues are helicity projectors of chirality -1. The gapless spectrum is charted by a dense-sweep census with computed Floquet–Weyl charges: 44 Weyl points in five tetrahedral orbits plus three unquantized-Berry-phase nodal lines, per-branch charges summing to $-10/+10$ with the two-branch total vanishing by conjugation pairing (no Floquet gap exists, so no zero-sum rule applies). Doubling by spatial inversion yields a Dirac fermion whose annealed-operator gap is exactly $2\arctan[q_m/(1-q_m)]$ in the density q_m of a fourth environment field, confirmed by gated measurement; the gap opens for the resonant principal quartet, while the off-resonant spectator species survive exactly massless. A media-mediated interaction with a continuous coupling is constructed and certified, with the free spectrum surviving to leading order. Finally we show that the annealed damping is an exactly mode-blind visibility decay—the annealed step operator factorizes as a scalar times a unitary—prove that within the quaternionic event class an isotropic cone forces this decay to be nonzero, with the exact bound $\kappa = \pi/(2\ln 2)$ on the node's phase advance per unit visibility decay (and exhibit the boundary of that class, where the bound fails), and show that at the fresh-read level the two-boundary (in–out) propagators form a one-parameter unitary family with the cone intact, within which the square-root weights are selected uniquely as the only medium-blind boundary.

I. INTRODUCTION

A probabilistic cellular automaton (PCA) is a classical statistical system: a lattice of bits, a probability distribution over bit configurations, and one local, invertible, homogeneous update rule applied in discrete time. Wetterich [1, 2] established that a class of such automata is *exactly* equivalent to fermionic quantum field theories in $1+1$ dimensions: the same wave function, density matrix, operators and expectation values, at every lattice size, with no approximation. The equivalence rests on three pillars: occupation bits map to fermions through a Grassmann functional integral; the step operator is a signed permutation, hence orthogonal; and a compatible complex structure turns orthogonal into unitary evolution. The construction in three or four spatial dimensions was left open—in Wetterich's own assessment such generalizations "do not seem to encounter problems of principle" but were "not yet achieved" [1]; the known four-dimensional route [3] transports Lorentz-invariant four-fermion composites rather than single fermions.

The program sits within a wider landscape. In the *unitary* quantum-cellular-automaton and quantum-walk tradition, discrete dynamics with postulated complex amplitudes yields Weyl and Dirac kinematics in one to three dimensions [6, 13–18], and Mlodinow and Brun have proven a no-go theorem for quantum-field limits of such automata above one dimension together with evasions that antisymmetrize distinguishable particles in two and three dimensions [19–21]; fermion doubling in these discrete settings remains an active subject [23]. The present work lives in the strictly harder *probabilistic* class, where amplitudes are not postulated but must emerge from classical statistics—the program of Refs. [1, 2, 8–11], with conceptual roots in 't Hooft's cellular-automaton interpretation [12]. The obstructions proven below (finite rays, Q-pinning) are properties of this probabilistic class; in the unitary setting, where complex amplitudes are postulated, the corresponding constraints do not arise in this form, which is precisely why the construction requires the dressing and coin mechanisms developed here.

This paper solves the three-dimensional problem for the free sector under the strictest contract: the propagating carrier is a single-fermion occupation number; the rule set (locality, invertibility with the unique-jump property, homogeneity, fermion-parity conservation per block, particle–hole symmetry, no hidden continuum input) is fixed at the outset and every construction below is verified against it mechanically.

The results divide into obstructions and constructions, and we state the evidence class of every claim: [PROVEN] for symbolic or algebraic proofs, [MACHINE-CHECKED] for exact finite verifications (tolerances quoted), [MEASURED] for measurements, which are always accompanied by a known-answer gate run through the identical analysis pipeline. The complete verification code is public [25].

Obstructions (Sec. III): ballistic one-particle transport in this class has a *finite* set of group velocities, so no three-dimensional cone is possible (Theorem 1); on the symmetric vacuum all transport speeds are rational, while a tuned vacuum density q provides a continuous, exactly stationary knob with the closed dressing law $v = 2|1-2q|$; Abelian position-periodic sign gauges relocate band features but cannot bend the factorized ("diamond") group-velocity surface into a cone; and two-component internal states admit isotropic nodes only at isolated fine-tuned points, every one frequency locked at $\omega_0 = \pi$ and without any dial (Theorem 2).

Constructions (Secs. IV–VIII): a four-component carried coin whose conversion blocks form the real quaternion representation of $\mathrm{Cl}(0,3)$ produces an exactly isotropic Weyl cone; the automaton realizing it (the *coincube*) is a composition of three manifestly legal layers whose fermionic lift is verified including all sign coherences; the step operator is exactly unitary with respect to an explicit complex structure; the local Grassmann action is extracted in exact rational arithmetic; a dense-sweep census with computed topological charges charts the full gapless spectrum; spatial-inversion doubling yields a Dirac fermion with an exact, continuously tunable mass for the resonant node quartet; and a media-mediated interaction with a continuous coupling is certified, with the free spectrum surviving.

Finally, Sec. IX addresses the one systematic cost of the construction. The dressed quasiparticle of the classical in-state observable is damped, and this is not a technical deficiency. The damping is exactly mode blind—the annealed step operator factorizes as a scalar times a unitary (Proposition 1)—and we prove that within the quaternionic event class an isotropic cone *forces* it to be nonzero, with the exact bound $\kappa = \pi/(2\ln 2)$ on the node's phase advance per unit visibility decay, correlated media included; outside that class the bound provably fails, and we exhibit the boundary. The two-boundary (in–out) propagators—legitimate objects in the same transfer-matrix formalism—form a one-parameter unitary family with the cone intact, and the square-root boundary is proven to be its unique medium-blind member.

Throughout, the lattice spacing and the duration of one full update cycle are unity; frequencies are per cycle and momenta are in lattice units.

II. SETUP

A. States, step operator, lifts

Configurations τ are assignments of occupation bits to species at the sites of a cubic lattice. The probabilistic information is a real wave function q_τ with $p_\tau = q_\tau^2$ [1]. One time step applies a deterministic, invertible map of configurations together with a sign gauge: the step operator $\hat{S}$ is a *signed permutation* of the configuration basis, hence real orthogonal. A rule is *legal* if it is a composition of layers, each of which is (i) local, (ii) a bijection on configurations with a unique image (the unique-jump property), (iii) homogeneous, and (iv) fermion-parity even on its support, so that a fermionic lift exists. The lift assigns the signs: occupation bits map to fermionic modes in a fixed Jordan–Wigner ordering, and each layer's signed permutation must equal the matrix of an explicit fermionic operator in the occupation basis. All lifts below are verified at this level, including two-particle sign coherence (Sec. IV).

B. Environment fields and dressed carriers

Besides the carrier species, the automata below contain autonomous binary *environment* fields: one per spatial axis (density q), plus one for the mass sector (q_m) and one for the interaction (g). Environment dynamics is free streaming by pair swaps; a Bernoulli product measure is exactly stationary. Carriers never modify the environment except through the explicitly constructed interaction layer of Sec. VIII. The vacuum is the product of the carrier vacuum with the Bernoulli environment measure: translation invariant, clustering, with a continuous parameter q that is the central dial of the construction.

III. OBSTRUCTIONS

Theorem 1 (Finite rays) *For any legal rule with finitely many species per site that conserves particle number in the one-particle sector—the sector with exactly one occupied bit across all species on an otherwise empty lattice—the one-particle Bloch operator is a monomial matrix for every momentum k. Consequently every dispersion branch is exactly linear in k, the set of group velocities is finite, and no three-dimensional cone $\omega = v|k|$ is possible ballistically.* [PROVEN]

The proof is elementary—a signed permutation with momentum phases has monomial matrix elements, and eigenvalues of monomial matrices are roots of single monomials—but the consequence is structural: the error is scale invariant, so no continuum limit removes it, and periodic cycles of rotated rules do not evade it because products of unique-jump operators are unique-jump. In one spatial dimension the light cone genuinely consists of two rays, which is why the 1+1-dimensional construction exists. An adversarial test suite (seven evasion strategies with re-seeded variants, plus reconstruction tests that fit the Bloch operator from real-space propagation rather than constructing it) accompanies the theorem in the repository. The same suite verifies how the construction of this paper relates to the theorem: at fixed media the coincube cycle is a signed permutation of the (site, channel) basis—the hypothesis holds per realization—while the measured, media-averaged operator is not monomial. The construction evades the theorem's hypothesis, not its conclusion.

Two further results fix the strategy. First, on the symmetric (density-$\frac{1}{2}$) vacuum the transport speeds of the autonomous-streaming subclass of conditional rules [2] are rational [PROVEN] (frozen-vacuum and fully coupled rules are outside this statement); the vacuum density q is the legal continuous parameter, with the closed dressing law, for the representative rule used throughout,

$$v(q) = 2\,|1-2q|, \tag{1}$$

proven as a two-class Markov-additive limit and verified against exact ring enumeration as polynomial identities in q [PROVEN]. Second, dressing alone yields a factorized walk whose group-velocity surface is the ℓ^1 ball (the *diamond*): slope ratios along (110) and (111) relative to (100) equal $\sqrt{2}$ and $\sqrt{3}$. Abelian position-periodic sign gauges (Kogut–Susskind axis phases [4, 22], per-stall signs, and their products) relocate the nodal points and modify residues but leave the ratios on the diamond, measured as $r_{110}/r_{111} = 1.40/1.68$ (base and stall gauges, identical spectra) and 1.46/1.82 (KS gauges) [MEASURED]—an early ungated instrument whose absolute anchor deviates 3.5–5.5% from the exact operator, quoted only for the on-diamond pattern; a carried non-Abelian structure is therefore necessary.

Theorem 2 (No protected two-component node) *Consider legal automata whose carrier has a two-dimensional internal space: per-axis cycles of monomial factors $(1-p)E_a + pC_a$ or $E_a[(1-p)\mathbb{1} + pC_a]$, with $E_a = \mathrm{diag}(e^{\pm ik_a})$, C_a a real signed permutation, and one or two (blocked) engagements per axis. (a) At every conversion weight $p \neq \frac{1}{2}$ the single-engagement cycles have no degenerate propagating pair at any momentum (rotation coins force $(1-p)^2 = p^2$; reflection coins never produce one; diagonal coins degenerate)* [PROVEN]. *(b) Isotropic two-component Weyl nodes exist only at* isolated fine-tuned points*: the single-origin placement at exactly $p = \frac{1}{2}$ ($\omega_0 = \pi$, $v = 1$, $\chi = -1$, $|\lambda_0| = 2^{-3/2}$, all forced, width $\Gamma = \frac{3}{2}\ln 2$ at the chord midpoint—the maximally damped point of Theorem 3)* [PROVEN]*; and the blocked two-origin cycles at numerically located tuned weights ($p^* = 0.3300$ additive, $p^* = 0.3204$ multiplicative), both likewise locked at $\omega_0 = \pi$ and heavily damped, with computed chiralities $\chi = -1$ and $\chi = +1$ respectively* [MACHINE-CHECKED]. *(c) At generic p the class has strictly positive anisotropy floors, established by an exhaustive sign/direction-table sweep* [MACHINE-CHECKED].

Two-component constructions therefore admit no *protected* relativistic carrier: every isotropic node is an isolated point in parameter space, its quasienergy locked at $\omega_0 = \pi$—no mass dial, no width dial, no vacuum-density dressing. The minimal internal space with a symmetry-protected node on an entire parameter family is four-dimensional and real, where the Clifford algebra $\mathrm{Cl}(0,3)$ has its quaternionic representation; that is the construction of the next section.

IV. THE COINCUBE AUTOMATON

A. Layers

The carrier has four species per site: a two-bit coin (b_1, b_2), channel index $c = 2b_1 + b_2$. Writing X, Z for the real Pauli matrices and XZ for their (antisymmetric) product, define the conversion triple and direction assignments

$$\begin{aligned} C_x &= XZ\otimes\mathbb{1}, & C_y &= Z\otimes XZ, & C_z &= -X\otimes XZ,\\ d_x &= \mathrm{diag}(Z\otimes\mathbb{1}), & d_y &= \mathrm{diag}(Z\otimes Z), & d_z &= \mathrm{diag}(\mathbb{1}\otimes Z). \end{aligned} \tag{2}$$

The C_a are pairwise anticommuting with $C_a^2 = -\mathbb{1}$: the real quaternion representation of $\mathrm{Cl}(0,3)$ [MACHINE-CHECKED]. Each full cycle applies, for each axis a and each of two block origins, three layers:

L2 (conversion). At every site, controlled on the axis-a environment bit at that site, the two channel pairs of C_a are swapped (a one-site block; two bit flips, parity even).

L1 (motion). Channel c translates by $d_a(c) \in \{\pm 1\}$ along axis a, unconditionally.

L3 (environment). The axis-a environment field pair-swap streams along axis $a+1$ (mod 3). This *cross-streaming* is mandatory: if the field streams along its own axis, a co-moving carrier re-reads its own bit, conversions arrive in nearly scalar bursts ($C_a^2 = -\mathbb{1}$), and the measured Weyl pair splits into one damped-real and one oscillating pole—an exceptional-point transition of the damped band structure in the sense of non-Hermitian topology [24]— and the node is destroyed [MEASURED].

Each layer is manifestly legal; the composition defines the automaton. The pair-swap streaming requires even linear size L (enforced in code); all statements below assume even L.

B. Fermionic lift and sign coherence

The lift of L2 is an environment-controlled pair of $\pm 90^\circ$ Givens rotations $G = \exp[\frac{\pi}{2}(a_c^\dagger a_{c'} - a_{c'}^\dagger a_c)]$ acting on same-site modes: a signed permutation of the Fock basis, parity-even, identity at empty control, with single-particle matrix equal to the corresponding signed-swap block of C_a and $G^2 = -\mathbb{1}$ on the one-particle sector; the quaternion sign structure *is* the fermionic rotation sign [MACHINE-CHECKED] (dense verification, all axes). Translations and environment swaps lift with the standard reordering parity. The composite cycle is sign coherent: in the segregated mode ordering (all carrier modes before all environment modes) the full lifted cycle is a signed permutation whose one-particle sector equals the stated rule and whose two-particle sector equals the exact antisymmetrized square $\Lambda^2 M_1$—verified in genuine three-dimensional geometry on all 5778 pair states at linear size $L = 3$, with mutation controls (deliberately corrupted sign rules fail the check) [MACHINE-CHECKED]. We record one instrument fact: at $L = 2$ the two sub-steps per axis make net translation crossings even, so $L = 2$ is blind to translation-sign errors; three-dimensional sign claims must be checked at $L \geq 3$.

V. FREE SPECTRUM: THE WEYL CONE

A. Annealed closed forms

With independent environment reads—the *annealed* idealization—the one-particle transfer per sub-step is $T_a(k) = E_a(k_a)\,[(1-q)\mathbb{1} + qC_a]$ with $E_a = \mathrm{diag}(e^{ik_a d_a})$, and the cycle operator is $U(k) = \prod_a T_a(k_a)^2$. Every exact spectral statement in this paper is about this annealed operator; the link to the streaming automaton is the gated measurement of Sec. V C, and the measured quenched node shifts quoted there quantify the difference between the two.

Proposition 1 (Scalar–unitary factorization) *Each conversion block C_a is real, antisymmetric and orthogonal, so $T_a(k)^\dagger T_a(k) = [(1-q)^2 + q^2]\,\mathbb{1}$ identically in k; hence*

$$U(k) = \rho^6\, V(k), \qquad \rho^2 = (1-q)^2 + q^2, \tag{3}$$

with $V(k)$ exactly unitary for every k. [PROVEN]

The proof is one line: $(\alpha\mathbb{1} + \beta C)^\dagger(\alpha\mathbb{1} + \beta C) = (\alpha^2 + \beta^2)\mathbb{1} + \alpha\beta(C + C^T) = (\alpha^2 + \beta^2)\mathbb{1}$; as a machine check, every eigenvalue modulus equals ρ^6 to 10^{-15} over random momenta at four values of q [MACHINE-CHECKED]. Two closed forms are immediate: the node modulus $|\lambda_0|^2 = \rho^{12} = [(1-q)^2 + q^2]^6$, and the exact periodicity $U(k + \pi e_a) = U(k)$, since $E_a(k_a + \pi) = -E_a(k_a)$ and the sign cancels in T_a^2 [PROVEN]. The linear splitting of the node doublet is proven symbolically to be exactly isotropic, identically in q, with speed $v(q)$ rising from $2/\sqrt{3}$ at $q \to 0$ to a maximum 1.207 near $q = 0.30$ ($v(0.08) = 1.1806$) [PROVEN]; numerically the splitting isotropy holds to 2×10^{-4} over 50 random directions after $h \to 0$ extrapolation [MACHINE-CHECKED].

B. The gapless census

The gapless set of the unitary factor $V(k) = U(k)/\rho^6$ is charted by a dense sweep of the reduced zone $[0, \pi)^3$: minimum pairwise eigenvalue distances on a 64^3 grid, Nelder–Mead refinement of every local minimum below 0.15 to machine precision, sphere-probe classification of each refined degeneracy (an isolated point keeps a finite gap in all directions; a line member has an antipodal zero pair), and orbit grouping under the spectral symmetry group. That group is determined empirically: of the 48 signed coordinate permutations, exactly the 12 proper rotations of the chiral tetrahedral group $T \cong A_4$ preserve the spectrum, and the 12 complementary operations (including $k \to -k$) map it to its conjugate—no odd permutation appears, reflecting the cyclic $x \to y \to z$ layer ordering. For every isolated node the chirality is *computed* as a Chern number: the Fukui–Hatsugai lattice Berry flux of the upper doublet member over an enclosing sphere, built from the spectral projectors of V (well defined by Proposition 1) and calibrated on $\exp(+ik\cdot\sigma)$; every quoted charge is an exact integer, stable under two sphere radii and two angular grids [MACHINE-CHECKED].

TABLE I. The gapless census of the coincube walk (chord weights, $q = 0.08$): five point orbits (44 isolated Weyl points), three nodal lines, and their triple junction. Quasienergies are +Im-branch values; charges are computed Chern numbers per node.

orbit	rep. k/π	ω/π	mult.	χ_+/χ_-
corner	$(0, 0, 0)$	0.1006	1	-1 / $+1$
half-points	$(\frac{1}{2}, 0, 0)$+perms	0.9221	3	$+1$ / -1
$\langle 111\rangle$ octet A	$(0.265, 0.265, 0.735)$	0.5142	8	-1 / $+1$
$\langle 111\rangle$ octet B	$(0.243, 0.243, 0.757)$	0.5652	8	$+1$ / -1
generic orbit	$(0.028, 0.401, 0.902)$	0.9479	24	-1 / $+1$
nodal lines	$(t, \frac{1}{2}, \frac{1}{2})$+perms	t-dep.	3	—
R junction	$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$	1	1	$U = -\rho^6\mathbb{1}$

Table I is the census. Beyond the corner and the half-points the zone carries 40 further Weyl points: two $\langle 111\rangle$ octets with members $(\pm t, \pm t, \pm t)$ mod π and a fully generic 24-point orbit. Point charges on the +Im branch sum to $S_+ = -1 + 3 + 4 - 4 - 12 = -10$ (and $S_- = +10$). Every other local minimum of the grid-basin search refines to a nonzero gap (smallest avoided crossing 0.019 at $q = 0.08$): to that resolution the list is complete. The same five-orbit topology with identical charges holds at chord $q = 0.15$ and at the two-boundary (arc) weights at $q = 0.08$—the extra species are not artifacts of the damping factorization—but it is not permanent in q: the generic orbit annihilates pairwise on the $\{k_a = 0\}$ planes at $q_c \in (0.292, 0.294)$, leaving a 20-point census with $S_+ = +2$ at $q = 0.30$ [MACHINE-CHECKED].

Two facts govern the charge balance. First, the eigenphases of $V(k)$ cover the entire circle—a 360-bin histogram over the zone has no empty bin—so V has no Floquet quasienergy gap anywhere and no Nielsen–Ninomiya [5] zero-sum rule applies to any branch or quasienergy window: $S_+ = -10$ violates nothing. The one exact global constraint is conjugation pairing: all layers are real, so $U(-k) = U(k)^*$ pairs every event (k, ω, χ) with $(-k, -\omega, -\chi)$—verified event by event—and the two-branch total vanishes identically. Second, the nodal lines carry no compensating charge: the loop Berry phase around a line is close to π but *not* quantized (0.954π–0.998π along the line; no protecting symmetry was identified), its winding along the full line vanishes within discretization accuracy, and a closed Gauss surface enclosing one line while avoiding the gapless set does not exist, since the three lines intersect at the R point [MACHINE-CHECKED].

All corner-local measurements below probe the doublet at $k = 0$, $\omega_0 = 0.1006\pi$. The nearest other gapless feature of any kind lies at torus distance 0.4134π from the corner (a generic-orbit member; the octets follow at 0.4216π and 0.4590π, the half-points at $\pi/2$, the nodal lines at 0.7071π), and the nearest spectator point node in quasienergy is 0.414π away, so corner-local fits at $|k| \lesssim 0.15$ are separated from every spectator by more than two decades in momentum. But the walk is a lattice fermion with a larger spectator census than a corner count suggests: its additional species are displaced, not absent [23].

C. Quenched measurement

On the streaming (quenched) environment the propagator is measured from the exact per-medium single-particle signed field, averaged over media. The instrument fits the full 4×4 one-cycle Bloch matrix per momentum from all four basis launches, pools symmetry-equivalent momenta at the level of characteristic-polynomial coefficients (basis invariant, hence unbiased) before rooting, probes momenta off the lattice grid inside the node's linear range, and extrapolates $\delta k \to 0$. Every production run carries an annealed known-answer row through the identical pipeline; a run whose gate row fails is discarded. Near-degenerate pole extraction is the dominant threat (eigenvalues of a noisy nearly defective matrix split as $\sqrt{\text{noise}}$), which this design removes.

Results (Fig. 1): each measured q carries its own annealed gate row, and the gate rows reproduce the exact operator (cubic-channel ratios within 10^{-3} of unity at $q = 0.08$ and within 6×10^{-3} at $q = 0.15$; off-symmetry channels within their own statistics; node modulus 0.622 vs 0.620 at $q = 0.08$), with the per-channel deviations recorded as systematics. On the quenched vacuum at $q = 0.08$ the slope ratios are $r_{110} = 1.001 \pm 0.002$,

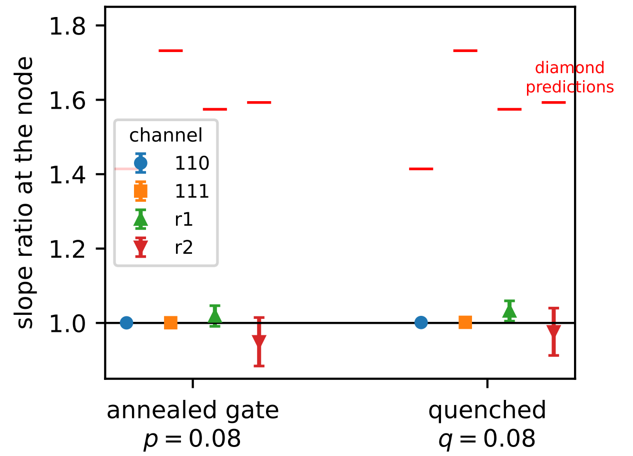

FIG. 1. Slope ratios at the node, gate and quenched rows at $q = 0.08$, against the factorized-transport (diamond) prediction. Error bars are leave-one-block-out jackknives over ten independent medium blocks with the gate-row systematic added in quadrature. [MEASURED]

$r_{111} = 1.002 \pm 0.002$, $r_{\mathrm{a}} = 1.032 \pm 0.027$ and $r_{\mathrm{b}} = 0.976 \pm 0.064$ for the two off-symmetry directions. The factorized-transport prediction is excluded in every channel: the weakest exclusion—by construction the off-symmetry channel with the fewest orbit members—sits 9.7 of its error bars from the diamond value (9.7σ under a Gaussian reading), and the cubic channels sit more than 200 error-bar widths away. A $q = 0.15$ run (own gate row passed) is committed alongside but is *excluded from the evidence*: its probe radii violate the pipeline's own linear-range rule (slope change 0.30 between probe radii), so neither its ratios nor its absolute scale are quoted here. Absolute speeds at $q = 0.08$ carry the full medium-ensemble uncertainty, $v = 1.03 \pm 0.18$, consistent with the exact 1.18. The quenched node parameters themselves shift from the annealed closed forms—$|\lambda_0|$ by $+2.8\%$ and ω_0 by $+8.6\%$ at $q = 0.08$—of which the gate row's own instrumental offsets account for $+0.3\%$ and $+2.3\%$; the remainder is a property of the quenched medium (re-read correlations), not of the instrument. [MEASURED]

D. Helicity residues

At the node the effective generators h_a on the doublet satisfy the exact Clifford algebra $\{h_a, h_b\} = 2v^2\delta_{ab}$; because the node frame of the damped (non-normal) operator is not unitary, the Pauli coefficients of h_a are complex with *bilinear* orthogonality $R^T R = v^2 \mathbb{1}$ ($R \in O(3, \mathbb{C})$), and $\det R = -v^3$ exactly: chirality $\chi = -1$ [MACHINE-CHECKED]. The split branches' spectral projectors equal the helicity projectors $\frac{1}{2}(\mathbb{1} \pm n \cdot \sigma)$, $n = Ru/\sqrt{(Ru)\cdot(Ru)}$. Measured on the quenched vacuum (the 26 directions of the cubic $\{100\}/\{110\}/\{111\}$ stars at each of the three—

by π-periodicity identical—corner representatives; gate row passed): chirality -1, stable under every jackknife resampling in both rows; pointwise helicity-direction agreement $|1 - n \cdot n_{\mathrm{exact}}|$ of mean 0.013 ± 0.004, with the gate row returning 0.015 ± 0.004 on the exactly isotropic operator—the instrument floor. The bilinear isotropy statistic (a maximum over the nine entries of $R^T R/s - \mathbb{1}$, hence positively biased) is reported only as an upper bound: the quenched row gives 0.075 ± 0.025 against the gate row's own 0.083 ± 0.036, so the measurement sits at the instrument's noise floor and bounds the node's bilinear anisotropy at $\lesssim 0.1$ [MEASURED]. (The residue projectors are rank one by construction of the spectral estimator, so "purity" is not reported as a measurement.) One convention is essential for eigenvector work: the $e^{-ik\cdot x}$ Fourier contraction estimates $U(-k) = \bar{U}(k)$, whose upper-half eigenvectors belong to the opposite-helicity conjugate doublet; residue measurements must use $e^{+ik\cdot x}$.

VI. EXACT QUANTUM MECHANICS

A. Complex structure

Let P be the lifted carrier particle–hole conjugation, the Majorana string $\prod_i (a_i + a_i^\dagger)$ over carrier modes. In the segregated ordering, P is real orthogonal with $P^2 = +1$ (carrier mode number $M = 4L^3 \equiv 0 \bmod 4$), and

$$[\hat{S}, P] = 0 \tag{4}$$

exactly, including all lift signs, as an operator identity on the full Fock space at every lattice size [PROVEN]: a string-factorization lemma reduces each layer's commutation with the Majorana string to a fixed finite block check (conversion and imprint blocks act on same-site modes, so their Jordan–Wigner spectators are block internal), and the translation lift contributes $\mathrm{sign}(\pi) = (-1)^{(L-1)L^2} = +1$ for every L (Appendix A). Independent sector certificates confirm the identity directly through the three-carrier sectors at $L = 2, 3$ (1.6×10^6 three-carrier states at $L = 3$), the $M-2$ sector by particle–hole duality, and one- and two-particle sectors at $L = 4$, over eight environment draws [MACHINE-CHECKED]. With the grading $\eta = \mathrm{sign}(N_c - M/2)$—conserved by every layer, odd under P, and size independent away from half filling—define

$$K = \mathrm{diag}(\eta), \qquad I = P\,\mathrm{diag}(\eta). \tag{5}$$

Then $K^2 = 1$, $I^2 = -1$, $\{K, I\} = 0$, I antisymmetric, and $[\hat{S}, I] = 0$; with multiplication $(\alpha + i\beta)q := \alpha q + \beta I q$ and the induced Hermitian form, $\hat{S}$ is a unitary operator on a complex Hilbert space [1]: the automaton's quantum mechanics is exact [MACHINE-CHECKED]. On the substrate the construction extends to the half-filled sector by an explicit orbit-wise grading; in three dimensions no size-independent grading of the half-filled sector was found and the question is left open. Away from half filling, all carrier sectors are covered at every size, by proof and by direct verification.

B. Bridge to the spectral analysis

The complex structure above is global (a Majorana string), and the spectral results of Sec. V use the ordinary Fourier i. These are the same quantum mechanics, by an explicit intertwiner: the map $\Phi(u + iv) = u \oplus (-Pv)$ is a complex-linear isomorphism from the complexified one-particle sector with the Fourier i onto the $(V_1 \oplus V_{M-1}, I)$ eigensector of the automaton's Hilbert space, intertwining the dynamics (this is exactly $[\hat{S}, P] = 0$ restricted to the sector) and the Hermitian forms. Consequently the measured propagator *is* an I-Hermitian matrix element, every $\pm\omega$ conjugate pair of the Fourier analysis appears once as a particle and once as an antiparticle in the (K, I) picture, and the chirality bookkeeping of Sec. V is the particle/antiparticle structure of a single Weyl node. Two scope remarks are owed. The complex structure is nonlocal—a Majorana string mapping k-particle to $(M-k)$-particle sectors—so no local occupation observable commutes with I; the identification of measured objects with I-Hermitian matrix elements is established here for the one-particle sector, and multi-carrier correlators (such as the C_2 certificates of Sec. VIII) are not re-expressed in the I picture. Machine-checked in three dimensions at $L = 2$ (one environment draw, 20 random vectors): the intertwining identities hold exactly and the form isometry to 3×10^{-15} [PROVEN], [MACHINE-CHECKED]. What remains genuinely distinct is the damping: the in-state reduced dynamics is a contraction (the chord), and only the two-boundary amplitudes of Sec. IX are unitary (the arc); the bridge identifies the kinematics, not the dissipative structure.

C. Grassmann action

The local factors and actions of all blocks are extracted mechanically in exact rational arithmetic by the pipeline of Ref. [1], Eqs. (51)–(58), validated term by term on the interaction block of that reference[1]. For the coin-cube conversion blocks the physical (Givens-lift) gauge yields an axis-universal action: 40 terms with identical degree structure for all three axes; the quadratic

[1] The validation surfaced one typographical error in Ref. [2]: the coefficient of the top monomial in the closed-form interaction action, Eq. (38) there (source label CS9; equation number confirmed by compiling the arXiv source), must be -2, not -1; the printed value fails the defining relation $e^{-L} = K$ at that single monomial.

part is the signed-swap bilinear of C_a plus the environment transport term, and the environment control appears as quartic and higher couplings between the channel bilinears and $e'\tilde{e}$ [MACHINE-CHECKED]. Axis universality is a property of the physical gauge specifically: the bare (all-plus sign) gauge yields axis-dependent term counts (49/42/51)—discarding the lift signs breaks the relabeling equivalence, a gauge artifact asserted as such [MACHINE-CHECKED].

VII. MASS: INVERSION DOUBLING

Doubling the coin with a mass bit b_m whose value *reverses all direction assignments*, $d_a^{(8)} = d_a \otimes \mathrm{diag}(1,-1)$, with conversions blind to b_m, places two opposite-chirality copies of the node at the same momentum and frequency (the b_m=1 sector is the spatial-inversion image; sector chiralities ∓ 1 verified) [MACHINE-CHECKED]. The alternative doubling $C_a \otimes Z$ fails for cause: $-C_a$ generates the opposite triple orientation, which is the *other* corner frequency family, so the two sectors share no node. A mass layer $M = (1-q_m)\mathbb{1} + q_m C_m$ with $C_m = \mathbb{1}_4 \otimes XZ$ (a b_m-flipping controlled Givens, driven by a fourth environment field of density q_m) couples the chiralities. Two operator identities carry the sector [PROVEN]: $U(k) = (\mathbb{1} \otimes R_m)[U_4(k) \oplus U_4(-k)]$ and, at the node, $U(0) = U_4(0) \otimes M_2$ with M_2 a scaled rotation of the mass bit; hence the gap is *exactly*

$$2m \; = \; 2\arctan\frac{q_m}{1-q_m}, \tag{6}$$

independent of q, with the multiplet center unshifted; frequencies add angles as in a telegraph process, and $m = q_m + O(q_m^2)$ recovers the Kac form [7]. On the node space the generators obey the full Dirac–Clifford algebra $\{h_a, h_b\} = 2v^2\delta_{ab}$, $\{h_a, \beta\} = 0$, $\beta^2 = 1$, identically in q [PROVEN], so $\omega = \omega_c \pm \sqrt{m^2 + v^2k^2}$ at leading order with the exact finite-k composition law $\cos(\omega - \omega_c) = \cos m \, \cos\varphi_{\mathrm{massless}}(k)$. No real Dirac mass exists at four components: the anticommutant of the quaternionic triple $\{C_a\}$ in $M_4(\mathbb{R})$ is zero-dimensional [MACHINE-CHECKED] (for a $\mathrm{Cl}(3,0)$ triple with squares $+1$ the anticommutant is two-dimensional with strictly negative squares— wrong-sign masses only); eight components are minimal, as realized here.

The census of Sec. V B has a sharp massive fate, charted by the same sweep at $q \in \{0.08, 0.15\}$, $q_m = 0.05$ [MACHINE-CHECKED]. The mass layer breaks all axis permutations (exactly the 8 diagonal sign flips survive as spectral symmetries) and makes the spectrum self-conjugate at every k, so both quasienergy branches live at the same momenta. The *resonant principal quartet*—the corner and the three half-points, the self-conjugate nodes with $2k \equiv 0 \pmod{\pi}$, where $\omega \equiv -\omega$—is gapped by exactly $2\arctan[q_m/(1-q_m)]$ (two exactly two-fold levels split by $2m$ to 10^{-15}). *Every one of the 40 extra Weyl points survives exactly ungapped*: for each, the original doublet at ω and its inversion partner at $-\omega$ re-close at a common point (coinciding to 6×10^{-14}, displaced below 0.005π from the massless location) with unchanged integer charges. The mechanism is off-resonance: inversion doubling places the partner at the conjugate quasienergy $-\omega \neq \omega$, and the mass layer gaps only resonant sector pairs. Survivor charges sum to zero per branch. The massive model also adds structures with no massless counterpart: degeneracy curves confined to the $\{k_a \in \{0, \pi/2\}\}$ mirror planes (where the intersector mass matrix element vanishes by symmetry), isolated Weyl nodes pinned at the resonant quasienergies $\omega \in \{0, \pi\}$ (16 events at $q = 0.08$, charges summing to zero), and a 12-line edge network that remains exactly degenerate, with the original line doublet split only weakly ($2m/9.8$ at $(0.7, \pi/2, \pi/2)$, $q = 0.08$), leaving subgap remnants. The massive model is therefore a Dirac quartet embedded in a spectator population that stays massless.

Measured on the quenched vacuum with the gated eight-launch instrument (Fig. 2): the gap is $m = 0.0576 \pm 0.0058\,(\mathrm{stat}) \pm 0.0150\,(\mathrm{syst})$ against the closed form 0.0526—the systematic being the gate row's own pair-splitting bias on the exactly known operator, measured with the identical estimator, which dominates the error budget—and the multiplet center is $\omega_c = 0.3190 \pm 0.0054$. The dispersion lies on the exact operator branches with maximum pulls of 1.2 (quenched row) and 1.7 (gate row) jackknife errors over the nine (family, δk) points—machine-asserted from the committed run output; the points share media blocks and the center estimate, so the worst-pull statistic is quoted, not a χ^2—with the family ordering of the curvatures that the exact model prescribes [MEASURED]. At this accuracy the instrument has no power to resolve quenched-medium shifts of the size seen in the node parameters (Sec. V C).

VIII. INTERACTION

For fixed media the multi-carrier sector is exactly determinantal (Sec. IV), so genuine interaction requires *back-reaction.* The imprint layer L4 acts once per cycle: at sites where an autonomous four-valued field fires (pair (xy), (yz) or (zx) with probability $g/3$ each) *and* the site's carrier fermion parity is odd, the selected two environment species at that site are swapped. The imprint field is autonomous in the same sense as the media: its randomness lives in the initial product measure only (value 0 with probability $1-g$; the four values are carried as two bits), and it evolves by legal deterministic pair-swap streaming with the axis rotating per cycle, lifting with the standard reordering parity like every environment layer. A per-cycle resampled field would be a stochastic rule—an annealed approximation, not an automaton—and is not used. Three design facts are load

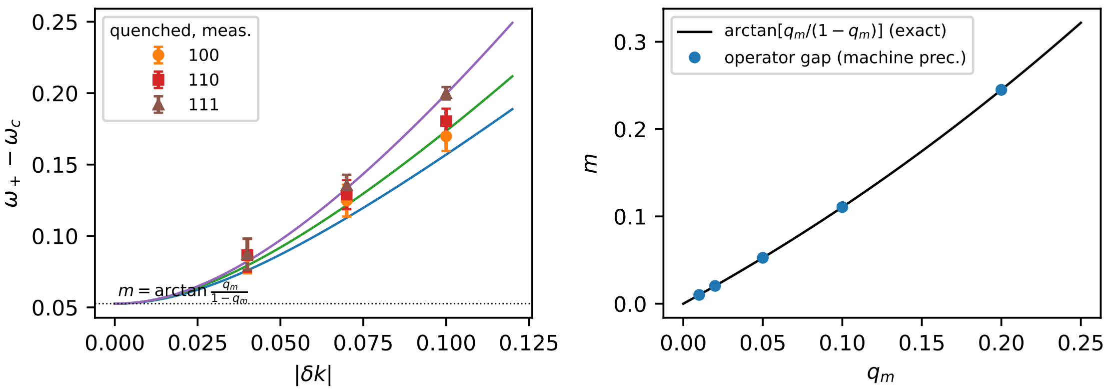

FIG. 2. Left: measured quenched massive dispersion (markers, $q = 0.08$, $q_m = 0.05$) on the exact operator branches (lines); the family ordering of the curvatures is the exact model's. Right: the operator gap against the closed form, Eq. (6), agreeing to machine precision. [MEASURED], [PROVEN]

bearing. (i) A fixed swap pair breaks cubic symmetry at $O(g)$ (anisotropic node corrections at 10σ) [MEASURED]; the rotating triple is protected by symmetry. The interacting quenched ensemble—media, imprint field, and the layered dynamics including L4—is exactly covariant under the tetrahedral rotation group A_4: cyclic axis rotations with a channel relabeling W (the accompanying layer-order shift is a similarity, being a cyclic rotation of an ordered product), and double axis reversals with a commutant relabeling V and bond-center reflections, verified quenched per mode *including* the imprint backreaction; single-axis reversals are provably not symmetries (no signed relabeling exists among all 384). A_4 acts irreducibly on $\mathbb{R}^3$ and admits no invariant vector and no non-scalar invariant symmetric rank-2 tensor, so at *every* order in g the node's velocity and damping tensors are scalar: the cone stays isotropic and undisplaced, with anisotropy possible only at rank ≥ 4 in δk, exactly as at $g = 0$ [PROVEN], [MACHINE-CHECKED]. (C_3 covariance alone would not suffice: $\sum_{a\neq b} k_a k_b$ is C_3-invariant and anisotropic.) (ii) Parity control (not occupancy) is particle–hole even, so the complex structure of Sec. VI survives at $g > 0$ unchanged [MACHINE-CHECKED]. (iii) The lift gauge of the imprint is *physical*: the permutation lift ($|11\rangle \to -|11\rangle$) and the Givens lift differ by a parity-controlled sign with identical classical dynamics but per-event coherent deficits $2q^2$ versus $2q(1-q)$; we fix the permutation lift (minimal decoherence) [MACHINE-CHECKED].

Certificates. The connected two-particle correlator $C_2 = G_2 - \det G_1$ is nonzero already at $g = 0$—the model is an interacting carrier–environment theory at the correlator level, with free-fermion structure only per medium realization—so the interaction certificate is differential: $\Delta C_2(g) = C_2(g) - C_2(0)$ vanishes identically at $g = 0$ and switches on linearly, $\Delta C_2/g \simeq 0.35$ on the exact substrate (a one-dimensional ring with a fixed swap pair and a static imprint field—an interaction certificate, not the production three-dimensional layer) [MACHINE-CHECKED]. A control-variant experiment (occupancy-controlled imprint, used only as an instrument control) separates the parity-blocking contact artifact: 2–3% of the signal [MACHINE-CHECKED].

Survival. The parity control makes the vacuum imprint-transparent, so the one-sided ensemble amplitude stays closed: $U_g(k) = (1 - 2gq^2)\, U(k)$ at annealed order—the cone, residues and mass are untouched, at a damping cost $\Gamma_{\text{int}} = 2q^2 g$ per cycle ($\sim 0.3\%$ of the free width at $q = 0.08$, $g = 0.1$) [PROVEN]. The identity that anchors the instrument: the media-ensemble walker average *equals* the canonical coherent-vacuum correlator exactly [MACHINE-CHECKED]. Quenched three-dimensional measurement with paired common-noise walkers (the instrument passes an exact known-answer gate: at $g = 0$ the paired branches agree bit for bit): the law is confirmed at the first cycle to 7×10^{-4} absolute (worst coupling, $g = 0.6$), with first-cycle momentum spread below 10^{-4} across the node region, so isotropy survives interaction by measurement as well; over six cycles the paired ratio stays momentum independent to $\leq 1.2\%$ wherever the amplitude gate retains all six probe momenta (at one intermediate cycle the gate retains a single momentum and k-independence is not claimed there); beyond leading order the re-read (recross) channel produces excursions on *both* sides of the law (within $\approx 3\%$ of it through six cycles), and the asymptotic interaction damping rate is left open [MEASURED].

IX. DAMPING: A THEOREM AND ITS UNITARY COMPLETION

The dressed quasiparticle of the classical in-state observable is damped: $\Gamma(q) = -3\ln[(1-q)^2 + q^2]$ exactly, with quality factor $Q = \omega_0/\Gamma$ between 0.60 and 0.90 across the working range, *not* improving as $q \to 0$ inside the relativistic window (both ω_0 and Γ scale with

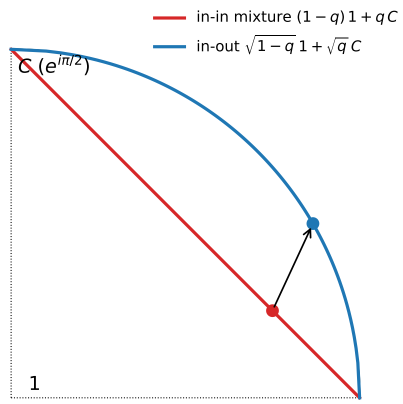

FIG. 3. Geometry of Theorem 3. Probabilistic (in-state) mixtures of the identity and a quaternion unit lie on the chord (damped interior); the two-boundary weights lie on the unitary arc. The isotropic node splitting is a property of the whole family $E_a(\alpha\mathbb{1} + \beta C_a)$, for all weights.

q). Proposition 1 states precisely what kind of damping this is: at annealed order $U(k) = \rho^6 V(k)$ with $V(k)$ unitary, so $\Gamma = -3\ln\rho^2$ is a mode-blind, k-independent *interference-visibility* decay—every coherent mode decays at the same rate relative to the conserved probability sector, whose Perron eigenvalue in the doubled description remains exactly 1. It is not a broadening of the node relative to any other mode. Nor is it a removable normalization: the in-state normalization is fixed by probability conservation, and the visibility-to-probability ratio is physical. All *mode-selective* decoherence resides in the quenched re-read corrections, quantified at the end of this section. That the visibility decay cannot be tuned away is structural:

Theorem 3 (Q-pinning, quaternionic event class) *Let every conversion event lie in the quaternionic set* $\mathbb{R}\cdot Q_8 = \{\pm\mathbb{1}, \pm C_x, \pm C_y, \pm C_z\}$. *Then for any product of convex mixtures of events—non-commuting factors and arbitrarily correlated weights included—every eigenvalue* λ *of the composite obeys* $\operatorname{dist}(\arg\lambda, (\pi/2)\mathbb{Z}) \le \kappa\,(-\ln|\lambda|)$ *with* $\kappa = \pi/(2\ln 2)$ *exactly. Moreover, under three further hypotheses—(i) events in* $\mathbb{R}\cdot Q_8$ *controlled by per-engagement reads of binary media; (ii) the coincube read geometry, in which the two same-axis engagements read lattice-adjacent sites with both offsets realized across coin branches; (iii) a deterministic environment rule with randomness confined to the initial measure—an isotropic node forces* $\Gamma > 0$*: node unitarity requires an almost-surely deterministic conversion word; among media factorizing over axis lines, the media with deterministic read-window counts are exactly the per-line crystal products, and every member destroys the cone.* [PROVEN]

The restriction to the quaternionic class is necessary, not stylistic: for transposition events on three or more modes (eigenvalue angles $\{0, \pi\}$) products of mixtures develop three-cycle eigenvalues near $e^{2\pi i/3}$, and the phase-per-decay ratio is unbounded (4.95 at mixing weight $\epsilon = 0.05$, 131 at $\epsilon = 0.002$); for event sets with finest eigenvalue angle $2\pi/\ell$ the constant grows as $\kappa(\ell) \to 2\ell/\pi$ [MACHINE-CHECKED]. The bound is a statement about the quaternionic class, degrading linearly with coin refinement. In these terms the theorem reads: no legal in-state automaton of this class can hold its node quasienergy farther than $\kappa\Gamma$ from $(\pi/2)\mathbb{Z}$. The bound governs the node—momenta where the transport phases vanish, so the composite stays inside the group algebra; at generic k the momentum factors $E_a(k)$ leave the event class and the ratio exceeds κ already on the coincube operator (reaching 6.06 at $q = 0.02$, $k = 0.4\,e_x$) [MACHINE-CHECKED]. And "an isotropic node forces $\Gamma > 0$" asserts a nonzero uniform visibility decay, not a node width.

The media hypothesis of the second statement is likewise scoped: for general three-dimensional translation-invariant media the deterministic-count classification is open—a family staggered along the read axis with an independent phase per transverse line lies outside the crystal list. It does not rescue the cone. Cross-streaming makes a carrier's two same-axis reads sample *adjacent* transverse lines, so per-line determinism does not give ensemble determinism: for independent phases of density r the ensemble one-cycle operator is damped by exactly $\Gamma = -3\ln[r^2 + (1-r)^2]$ per cycle, vanishing only at the crystal limits $r \in \{0, 1\}$; and an exhaustive supercell construction (all 4096 members of the family on the 2^3 supercell, each exactly unitary and closed under the model's streaming) finds every one of its 11 864 degenerate band groups failing an isotropic-pair witness at the folded corner momenta—a witness that does detect the unitary-arc cone when fed one. Interior supercell momenta are not exhaustively excluded. [MACHINE-CHECKED]

The idealized ordered-dimer environment does achieve the commutator-limited decoherence $\Gamma = k^2/2$ per event—the branch-remerging mechanism is real—but its velocity fan collapses to the diamond: within the legal class, cone and in-state unitarity are mutually exclusive.

The completion is the two-boundary object. The isotropic node splitting holds for the whole family $E_a(\alpha\mathbb{1} + \beta C_a)$, and by the identity of Proposition 1, $T^\dagger T = (\alpha^2 + \beta^2)\mathbb{1}$ for every real weight pair (the two-boundary walk is the $\alpha^2 + \beta^2 = 1$ member of the same family): *every* product final boundary (w_0, w_1) yields zero relative damping and an isotropic node, landing on the unitary arc at $s^2 = w_1^2 q/[w_0^2(1-q) + w_1^2 q]$ [PROVEN]. The completions thus form a one-parameter family; in-state weights $(1-q, q)$ lie on the damped chord, and the square-root point $(\sqrt{1-q}, \sqrt{q})$ (Fig. 3) is selected uniquely as the only *medium-blind* boundary—the only choice invariant under flipping the environment bits ($w_0^2 = w_1^2$) [PROVEN]. Numerically: all $|\lambda| = 1$ to 4×10^{-16} with slope isotropy asserted below 10^{-3} and ω_0 tunable at the $\sqrt{q}$ scale [MACHINE-CHECKED]. In quenched three dimensions the fresh-read law is exact at the first cycle (coherent path enumeration, no sampling), with growing re-read corrections thereafter (11–24% by three cycles at $q = 0.08$); the long-time pole structure of the quenched two-boundary object is open. [MACHINE-CHECKED]

X. DISCUSSION

Under the strictest single-fermion contract, the free-fermion sector of the three-dimensional problem posed in Ref. [1] is solved: a legal probabilistic cellular automaton whose dynamics is exactly unitary quantum mechanics and whose single-carrier excitation on a clustering product vacuum is, by gated measurement, an isotropic Weyl fermion—dispersion and spinor residues—extendable to a massive Dirac quartet with an exact tunable mass, and to a certified media-mediated interaction under which the free spectrum survives at leading order. Every closed form is verified against the operators it describes, and every measurement runs behind a known-answer gate. The solution is a lattice fermion with a computed spectator census: the sweep of Sec. V B places 43 further Weyl points and three nodal lines at momentum distance $\geq 0.41\pi$ from the working node, and the massive model gaps only the resonant quartet among them.

The precise open problems are: (i) the half-filled carrier sector of the complex-structure construction in three dimensions; (ii) the long-time behavior of the quenched two-boundary propagator (re-read resummation); (iii) the continuum limit and the fate of Lorentz symmetry beyond the leading cone, including interaction corrections beyond the scalar law; and (iv) multi-carrier physics beyond the contact terms already present in the extracted quartic action; and (v) the spectator population of the massive census—the exactly massless off-resonant Weyl species, the pinned nodes, and the sub-gap nodal-line remnants—and whether a legal layer can gap it. The damping of the in-state quasiparticle is not on this list: it is an exact, mode-blind prediction of the construction (Proposition 1), bounded by Theorem 3, and absent from the two-boundary amplitude.

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Appendix A: Proofs

1. Theorem 1 (finite rays)

Let the rule conserve particle number in the one-particle sector. Its restriction to that sector is a signed permutation of the species basis composed with site shifts, so the Bloch operator has matrix elements $S(k)_{\alpha\beta} = s_{\alpha\beta} e^{ik\cdot d_{\alpha\beta}}$ with, by the unique-jump property, exactly one nonzero entry in each row and column: a monomial matrix. Its characteristic polynomial factorizes over the cycles of the underlying permutation; a cycle of length ℓ, accumulated displacement D and sign σ contributes $\lambda^\ell = \sigma\, e^{ik\cdot D}$, i.e. $\omega_m(k) = (k\cdot D + \arg\sigma + 2\pi m)/\ell$: every branch is exactly linear with group velocity D/ℓ, drawn from the finite set of cycle data. A cone $\omega = v|k|$ requires a two-sphere of group velocities. The deviation of any finite ray set from a sphere is scale invariant, so no continuum limit removes it; and a periodic cycle of rotated rules is again unique-jump (products of signed permutations are signed permutations), so the theorem applies to the composite rule. □

2. Theorem 2 (two-component states)

Three ingredients. (i) *Algebraic ceiling.* On $\mathfrak{sl}_2(\mathbb{R})$ every element satisfies $X^2 = -\det(X)\mathbb{1}$ and anticommuting elements are orthogonal in the polarization of det; in the basis $\{X, Z, XZ\}$ the squares are $+\mathbb{1}, +\mathbb{1}, -\mathbb{1}$, so the form has signature $(2, 1)$ and no real two-dimensional representation of $\mathrm{Cl}(3, 0)$ exists. (ii) *Generic weights.* A symbolic orthogonality lemma for the mixture families shows that a degenerate propagating pair at any momentum requires $(1-p)^2 = p^2$ for rotation coins, is impossible for reflection coins, and degenerates for diagonal coins: a linear node forces $U(k_0) = \lambda_0\mathbb{1}$, the splitting generators are conjugated involutions with automatically scalar anticommutators (2×2 Cayley–Hamilton), and the required bilinear orthogonality gives $T_{00}T_{11} = -T_{01}T_{10}$. For the blocked two-origin cycles the exclusion is exhaustive in the tables and heuristic in momentum: all 512 sign/direction designs per (placement, engagement, p) sector are enumerated, with $p \in \{0.15, 0.25, 0.35, 0.5\}$ and 14-start continuous momentum solving, and every design carries an explicit certificate—a candidate node entering the sector's anisotropy floor, a degenerate-only scalar point, or no scalar point at all, the last certified by a minimum scalar residual ≥ 0.074 against the 10^{-9} solver tolerance of solvable designs. The floors are claimed at the sampled grid values of p only; between them they dip toward zero on approach to the tuned p^*, which is the content of part (iii), not a violation. (iii) *The fine-tuned points.* At the singular weight $p = \frac{1}{2}$ the single-origin placement has $U(k_0)$ scalar at $k_0 = (\frac{3\pi}{4}, \frac{3\pi}{4}, -\frac{\pi}{4})$ (and its symmetry images), and the three effective generators form an exact complex Clifford triple, giving $\omega = \pi \pm |\delta k|$: an isotropic node with every parameter frozen and $\Gamma = \frac{3}{2}\ln 2$ at the chord midpoint. In the blocked placements, branch continuation in p (warm-started exact scalar solves with probe-radius-extrapolated anisotropy) locates codimension-one transversal zeros of the single remaining Gram condition at $p^* = 0.3300$ (additive placement, $k_0/\pi = \pm(0.916, 0.916, -0.916)$, $|\lambda_0| = 0.1736$, $\chi = -1$) and $p^* = 0.3204$ (multiplicative, $|\lambda_0| = 0.1799$, $\chi = +1$), both with $\omega_0 = \pi$: isolated tuned nodes, not parameter families. All parts are machine-checked, the sweep exhaustively. □

3. Particle–hole equivariance, $[\hat{S}, P] = 0$

Lemma (string factorization). Let B be a parity-even operator with support T, and T_c the carrier modes in T. The Majorana factors of P anticommute pairwise, so sorting gives $P = \sigma\, P_{T_c} P_{T_c^\complement}$ with a fixed reordering sign $\sigma = \pm 1$; a parity-even B commutes with every Majorana off its support, hence with $P_{T_c^\complement}$, and $[B, P] = \sigma\,[B, P_{T_c}]\, P_{T_c^\complement}$: the block-local check suffices.

Layers. L2 acts, per fired site, on the four same-site carrier modes (plus the control); the stored Jordan–Wigner spectators lie inside the contiguous block, so $[\mathrm{L2}, P] = 0$ reduces to one finite block check per axis, independent of L. L1 is a canonical permutation lift, for which $U_\pi P U_\pi^{-1} = \mathrm{sign}(\pi)\, P$; per channel the axis-$a$ shift is L^2 disjoint L-cycles, so $\mathrm{sign}(\pi) = (-1)^{(L-1)L^2} = +1$ for every L (even L: L^2 even; odd L: $L-1$ even). L3 is supported on environment modes and parity even, so it commutes with every carrier Majorana. L4's carrier part is a parity projector and its environment part a string-free adjacent swap, block-checked in both lift gauges. Size independence is structural: the same finite blocks recur at every site of every lattice, so the identity holds on the full Fock space at every L. □

4. The mass identities, Eq. (6)

First, the factorization $U(k) = (\mathbb{1} \otimes R_m)[U_4(k) \oplus U_4(-k)]$ with $R_m = (1 - q_m)\mathbb{1} + q_m XZ$: (1) $d^{(8)} = d \otimes \mathrm{diag}(1, -1)$, so every transport phase is block diagonal in b_m with opposite momentum sign in the two blocks, $E_8(k) = E_4(k) \oplus E_4(-k)$; (2) $C_8 = C_4 \otimes \mathbb{1}$, so conversions are block diagonal and identical in the two blocks; hence the pre-mass cycle is $U_4(k) \oplus U_4(-k)$ and the once-per-cycle mass layer is $\mathbb{1}_4 \otimes R_m$ (machine-checked symbolically in the full vector momentum). Second, the gap: at $k = 0$ the momentum factors are the identity, so the direction reversal that distinguishes the two b_m sectors is invisible and the axis layers act as $U_4(0) \otimes \mathbb{1}_2$. The mass layer acts on the b_m factor alone: $M_2 = (1 - q_m)\mathbb{1} + q_m XZ$, a real multiple of a rotation with arg-eigenvalues $\pm\arctan[q_m/(1 - q_m)]$ and modulus $\sqrt{(1 - q_m)^2 + q_m^2}$. Hence $U(0) = U_4(0) \otimes M_2$: the spectrum is the product set, frequencies add angles, the multiplet center is unshifted, and the gap is exactly $2\arctan[q_m/(1 - q_m)]$, independent of q. □

5. The imprint amplitude law

In the one-particle sector the carrier's site has odd parity, so the imprint fires iff $\iota \neq 0$, probability g. The permutation lift of the fired pair swap has sign table $|00\rangle \to +|00\rangle$, $|01\rangle \to +|10\rangle$, $|10\rangle \to +|01\rangle$, $|11\rangle \to -|11\rangle$ (the canonical transposition lift), and the swap preserves the Bernoulli pair measure, so the per-event coherent factor is the plain sign average $(1-q)^2 + 2q(1-q) - q^2 = 1 - 2q^2$, identical for the three pairs. One line of expectation: $U_g = (1 - g)U + g(1 - 2q^2)U = (1 - 2gq^2)U$ at annealed order. The Givens lift's table gives $1 - 2q(1 - q)$ instead: the law is lift-gauge dependent, the isotropy of the damping is not. □

6. Theorem 3 (Q-pinning)

Any product of convex mixtures of the event set is itself a convex mixture over the quaternion group Q_8 (the product measure is the convolution on the group, so correlations between events are automatically included). In the quaternion representation such a mixture is $T = w\mathbb{1} + xC_x + yC_y + zC_z$ with $|w| + \|(x, y, z)\|_1 \le 1$, and its eigenvalues are $w \pm i\|(x, y, z)\|_2$. Since $\|\cdot\|_2 \le \|\cdot\|_1$, every eigenvalue lies inside the convex hull of the chords between adjacent fourth roots of unity. On the extremal chord $1 \to i$, $\lambda(t) = (1 - t) + it$, the maximized quantity is $\mathrm{dist}(\arg\lambda, (\pi/2)\mathbb{Z})/(-\ln|\lambda|)$: by the square-hull reduction and the $\pi/2$ rotation symmetry of the event angles it attains its maximum $\pi/(2\ln 2)$ at the chord midpoint $t = \frac{1}{2}$, where the distance folds from $\arg\lambda$ to $\pi/2 - \arg\lambda$. (The unfolded ratio $\arg\lambda/(-\ln|\lambda|)$ diverges toward the chord endpoint and is maximized nowhere interior; a dense hull scan confirms no interior point exceeds κ.) This bounds $\mathrm{dist}(\arg\lambda, (\pi/2)\mathbb{Z}) \le \kappa\Gamma$ for arbitrary, non-commuting products—the full statement of the theorem. The group closure is the load-bearing hypothesis: outside $\mathbb{R} \cdot Q_8$ (already for transposition events on three modes) products of mixtures leave the span of the event set, three-cycle eigenvalues appear, and the measured phase-per-decay ratios grow without bound, as reported in the main text. For the second statement (under hypotheses (i)–(iii), the read-window structure being hypothesis (ii)): unitarity of the node eigenvalue requires the conversion count N (mod 4) along a read window to be deterministic, by $|\mathbb{E}[i^N]|^2 = (p_0 - p_2)^2 + (p_1 - p_3)^2 \le 1$ with equality only at a vertex; on rings, and on three-dimensional media factorizing over axis lines, the media with deterministic window counts are exactly the (per-line) crystals (exhaustively enumerated at $L = 6, 7, 8, 10$), with the non-factorizing phase-per-line family closed separately as described in the main text; and the deterministic-word cycles' spectra are computed explicitly and are never isotropic. Machine checks accompany every step, including randomized non-commuting products. □

Appendix B: Verification protocols

Every claim tagged [MACHINE-CHECKED] is an assertion in a runnable script; the tolerances quoted below are the assertion thresholds, and each script is self-checking (it fails loudly rather than reporting).

Layer lifts. The conversion layer is verified densely on the one-site Fock space (32 states): the controlled double Givens is a signed permutation of the basis, unitary, commutes with fermion parity, is the identity on the empty control sector, has single-particle matrix equal to the signed-swap block of C_a, satisfies $G^2 = -\mathbb{1}$ on the one-particle sector, and maps the doubly occupied pair to + itself (determinant one), for all three axes, to 10^{-12}.

Composite sign coherence. The full cycle is built as a signed permutation of the many-body basis from the stored per-layer sign rules (spectator Jordan–Wigner strings for conversions; inversion and wrap parity for translations), in the segregated mode ordering. For random environment configurations the zero-, one- and two-particle sectors are extracted and compared: the one-particle sector must equal the stated rule and the two-particle sector the antisymmetrized square $\Lambda^2 M_1$, exactly. This holds on the full Fock space of a ring substrate (2^{15} states) and in three-dimensional geometry at $L = 2$ (496 pair states) and $L = 3$ (5778 pair states). Mutation controls certify sensitivity: deleting the spectator string produces 8/496 mismatches; deleting translation parity produces 2772/5778 at $L = 3$ (counts are specific to the committed environment draw) and—the recorded caveat—0 at $L = 2$, where two sub-steps per axis make net crossings even.

Complex structure. P is applied as the Majorana string with its Jordan–Wigner signs; $[\hat{S}, P] = 0$ is checked as an identity of signed permutations (both orderings byte-identical), on the full substrate Fock space and, in three dimensions, on the sectors $\{0, 1, 2, M-2, M-1, M\}$ using particle–hole duality to reduce the deep sectors to one- and two-hole problems; per layer and for the full cycle. The unitarity of the complex picture is verified as invariance of the induced Hermitian form on random vectors to 10^{-9}.

Grassmann extraction. Local factors are built from the block tables over exact rationals; $e^{-L} = K$ is asserted exactly before any action is reported, and the inverse map $K \to (\text{perm}, \text{signs})$ must round-trip. The pipeline is validated term by term against the closed-form interaction action of Ref. [1].

Table II maps each principal claim to its verification script in Ref. [25].

TABLE II. Principal claims and their verification scripts (all self-asserting; in `scripts/` of Ref. [25] unless noted).

finite-ray tests	`tests/test_finite_ray_*`
$v(q)$ dressing law	`theory_linear_law`
Abelian-gauge diamond	`w2_scalar_gauges`
coin scans ($n = 2, 4$)	`w3_transfer_scan`
annealed cone	`w3_cone_verify`, `theory_real_cone`
lift certificates	`w3c_lift_check`, `w3c_composite_check`
3D sector certificates	`theory_3d_certs`
$[\hat{S}, P]$ equivariance proof	`theory_sp_proof`
quenched cone (gated)	`w3c_corner`
gapless census, charges	`census_sweep`
factorization; structural loci	`spectrum_census`
estimator positive controls	`w3c_positive_control`
two-component taxonomy (Thm 2)	`theory_redteam`
helicity residues	`w4_helicity_exact`, `w4_helicity`
complex structure	`e1_complex_structure`, `e1_interacting`
Grassmann actions	`e2_coincube_actions`
mass sector	`m8_mass_exact`, `m8_corner`, `theory_mass`
mass factorization	`theory_mass_identity`
massive dispersion pulls	`m8_pulls`
interaction	`i2_connected`, `i2_split`, `i3_quenched3d`
imprint-field dynamics	`theory_interaction`
Q-pinning; in–out	`theory_q_bound`, `a_inout`, `a_inout_3d`

Appendix C: Measurement pipeline

All [MEASURED] results use one instrument family. The object is the exact per-medium single-particle signed field (the single-carrier sector is linear in the field for fixed media, so this is walker-exact with infinite walkers), averaged over R media; the media-ensemble average provably equals the canonical coherent-vacuum correlator.

Bloch-matrix estimation. All n basis launches are evolved, giving the full $n \times n$ propagator series $G(t, k)$; the one-cycle operator is fit by least squares, $\hat{U}(k) = G_{t+1} G_t^+$ over a post-transient window. Momenta are probed off the lattice grid by direct phase contraction, with probe radii inside the node's measured linear range.

Pooling. Naive eigenvalue extraction near a two-fold node is $\sqrt{\text{noise}}$-ill-conditioned (a noisy nearly defective matrix splits its double eigenvalue as the square root of the perturbation); pooling is therefore performed on characteristic-polynomial coefficients over the symmetry orbit of measurement momenta—a basis-invariant, *linear* (hence unbiased) average—before rooting. Slopes use symmetric splittings at two step sizes with $\delta k \to 0$ Richardson extrapolation. Media are accumulated in independent blocks (ten for the cone instrument, eight for the massive instrument, six for the helicity instrument) and every final quantity (slope, ratio) is jackknifed at the end-of-pipeline level over leave-one-block-out samples; the annealed gate row's deviation from the exact operator is recorded per channel and added in quadrature as a systematic. Off-symmetry direction families are measured alongside the cubic stars, and the diamond-exclusion significance is computed in code for every channel, with the weakest channel quoted.

Gates. Every production run carries an annealed known-answer row through the identical pipeline; a failed gate discards the run. Amplitude gates exclude momenta near free-beat amplitude nodes for ratio estimators (shot-noise gating alone is insufficient there). Eigenvector work uses the $e^{+ik\cdot x}$ contraction (Sec. V). For interaction ratios, paired common-noise evolution (identical randomness with the imprint on and off) cancels the medium-ensemble variance, which is otherwise dominated by low-order conversion caustics. Gate tolerances are set to catch catastrophic estimator failure, not to certify accuracy: the cone-ratio gates use $\max(2\%, 3\sigma)$ of the gate row's own jackknife, the massive-gap gate allows 30% of the gap, and the node modulus gate 0.05 absolute. Accuracy statements never rest on gates; they rest on the quoted statistical errors and the recorded systematics, and the estimator's ability to resolve anisotropy is established by positive controls (a synthetic diamond-splitting truth and a legal broken-coin walk) run through the identical pipeline.

Recorded failure modes. The following estimator pathologies were each encountered, diagnosed and firewalled during this program, and are listed because they generically afflict lattice pole measurements: (i) $\sqrt{\text{noise}}$ splitting at near-degenerate poles; (ii) exceptional-point misidentification in damped operators (real double eigenvalues accepted as propagating nodes); (iii) through-origin fits across a packet's death into the noise floor; (iv) ratio blowup at beat-amplitude nodes; (v) probing outside the (renormalized) linear cone window; (vi) branch mixing under scalar single-pole estimators at two-fold degenerate nodes; (vii) one-dimensional co-streaming substrates, whose companion locking invalidates absolute calibrations that do transfer in three dimensions.

Appendix D: Closed forms

Table III collects the closed forms. Two remarks. The in–out transfer's uniform modulus $1/D$ per read is a global scalar (every path reads one bit per sub-step) and is absorbed into wave-function renormalization; the physical statement is the vanishing of all *relative* damping. The quality factor of the in-state quasiparticle, $Q = \omega_0/\Gamma$, rises from 0.60 to 0.90 over $q \in [0.02, 0.25]$; the quantity Theorem 3 bounds by κ is $\mathrm{dist}(\omega_0, (\pi/2)\mathbb{Z})/\Gamma$, which coincides with Q for $q \lesssim 0.2$ (where $\omega_0 < \pi/4$).

TABLE III. Exact expressions of the free and dressed spectra. All are verified against the operators they describe to at least 10^{-9}; $D = \sqrt{1-q} + \sqrt{q}$.

node modulus	$\lvert\lambda_0\rvert^2 = [(1-q)^2 + q^2]^6$
factorization	$U(k) = \rho^6\, V(k)$, $\rho^2 = (1-q)^2 + q^2$, V unitary
free width	$\Gamma(q) = -3\ln[(1-q)^2 + q^2]$
cone speed	$v(q)\!:\ 2/\sqrt{3} \to 1.207$ (max at $q \simeq 0.30$)
dressing law (1D)	$v = 2\lvert 1-2q\rvert$
mass gap	$2m = 2\arctan[q_m/(1-q_m)]$
mass-layer modulus	$\sqrt{(1-q_m)^2 + q_m^2}$ per cycle
massive dispersion	$\cos(\omega - \omega_c) = \cos m\, \cos\varphi(k)$
interaction law	$U_g(k) = (1 - 2gq^2)\, U(k)$
Q-pinning constant	$\kappa = \pi/(2\ln 2) = 2.2662$
in–out transfer	$E_a(k)\,[\sqrt{1-q}\,\mathbb{1} + \sqrt{q}\, C_a]/D,\ \ \lvert\lambda\rvert \equiv 1/D$

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